Differential Equations

1st-order Homogenous Diff eq.

If the diff eq. can be expressed in the form:

$$y' = \frac{f(x)}{g(y)}$$

We use <u>separation of variables</u>!

Approach:

1) Rearrange $\quad g(y)y' = f(x)$

$\boxed{\text{Leibniz}}$ $\Rightarrow \quad g(y)\frac{dy}{dx} = f(x)$

$\Rightarrow \quad g(y)\,dy = f(x)\,dx$

2) Take antiderivatives of both sides

$$\int g(y)\,dy = \int f(x)\,dx$$

$$\Rightarrow \quad G(y) = F(x) + C$$

3) Solve for y

Always get soln of form $y_0 e^{-ax}$

Problem $\boxed{4}$ The temperature deviation in a container satisfies:

$$\dot{T}(t) + \frac{1}{6} T(t) = 0 , \qquad T(0) = 12$$

$$\Rightarrow \quad y' = \frac{f(t)}{g(y)} \qquad \left| \begin{array}{l} g(y) = \frac{1}{T(t)} \\ f(t) = -\frac{1}{6} \end{array} \right.$$

ⓐ Find solution

1) Rearrange to get T terms on one side:

$$\frac{1}{T(t)} \dot{T}(t) = -\frac{1}{6}$$

Leibniz $\Rightarrow$ $$\frac{1}{T(t)} \frac{dT(t)}{dt} = -\frac{1}{6}$$

$$\Rightarrow \quad \frac{1}{T(t)} dT(t) = -\frac{1}{6} dt$$

2) Take the antiderivatives of both sides

$$\int \frac{1}{T(t)} dT(t) = \int -\frac{1}{6} dt$$

$$\ln |T(t)| = -\frac{1}{6} t + C$$

3) Solve for $T(t)$

$$e^{\ln |T(t)|} = e^{-t/6 + C}$$

$$\Rightarrow \quad T(t) = Ke^{-t/6}$$

(b) Determine the time constant

→ Use initial conditions $T(0) = 12$

$$T(0) = 12 = Ke^{0/6} = K$$

$$K = 12$$

(c) Determine the time (t) at which the deviation T has decreased to 5% of its initial value T(0).

$$T(0) = 12$$

$$5\% = 12(0.05)$$

→ Plug in and solve for t.

$$12(0.05) = 12e^{-t/6}$$

$$0.05 = e^{-t/6}$$

$$\ln(0.05) = -\frac{t}{6}$$

$$\ln(0.05) \cdot 6 = -t$$

$$t = -6\ln(0.05)$$ ✓

1st-order Forced (inhomogenous) Diff Eqs

Given $y' + f(x)\,y = g(x)$

1) Set the forcing term to zero $g(x) = 0$

2) Solve the homogenous diff eq.

$$y_h' + f(x)\,y_h = 0$$

$$\Rightarrow y_h' = -f(x)\,y_h$$

$$\Rightarrow \frac{1}{y_h} y_h' = -f(x)$$

$$\Rightarrow \frac{1}{y_h}\frac{dy_h}{dx} = -f(x)$$

$$\Rightarrow \frac{1}{y_h} dy_h = -f(x)\,dx$$

$$\Rightarrow \int \frac{1}{y_h} y_h = \int -f(x)\,dx$$

$$\Rightarrow \ln|y_h| = -\int f(x)\,dx + C$$

$$\Rightarrow \boxed{y_h = Ke^{-\int f(x)\,dx}}$$

3) Solve the forced diff eq. by <u>variation of constants</u>

$\Rightarrow$ Replace constant K from the homogenous soln with an arbitrary function $K(x)$:

$$y(x) = k(x) e^{-\int f(x)dx}$$

$$y'(x) = k'(x) e^{-\int f(x)dx} + k(x) e^{-\int f(x)dx}(-f(x))$$

$\Rightarrow$ Plug back into original diff eq:

$$y' + f(x) y = g(x)$$

$$\Rightarrow \underbrace{k'(x) e^{-\int f(x)dx} + k e^{-\int f(x)dx}(-f(x))}_{y'(x)} + \underbrace{k e^{-\int f(x)dx}(f(x))}_{f(x)\, y} = g(x)$$

Cancel

$$\Rightarrow k'(x) e^{-\int f(x)dx} = g(x)$$

4) Integrate both sides w.r.t x

$$\Rightarrow k'(x) = g(x) e^{\int f(x)dx}$$

Integrate $\Rightarrow$ $$k(x) = \int g(x) e^{\int f(x)dx} dx + C$$

5) Plug $k(x)$ back into our homogenous diff eq!

$$y_h(x) = k e^{-\int f(x)dx}$$

$$\Rightarrow y(x) = k(x) e^{-\int f(x)dx}$$

$$\Rightarrow \boxed{y(x) = \left[\int g(x) e^{\int f(x)dx} dx + C \right] e^{-\int f(x)dx}}$$

done!

5 The order backlog in a distribution center is described by

$$\dot{B}(t) + 0.5B(t) = 30 \qquad B(0) = 0$$

ⓐ Determine the complete soln.

1) set the forcing term to zero

$$\dot{B}(t) + 0.5B(t) = 0$$

2) Solve the homogenous diff eq.

$$B_h(t) = ke^{-0.5t}$$

3) Solve the forced diff eq. by variation of the constants.

3.1) $B(t) = k(t)e^{-0.5t}$

3.2) $B'(t) = k'(t)e^{-0.5t} + k(t)e^{-0.5t}(-0.5)$

3.3) Plug $B(t)$ and $B'(t)$ back into original

$$\dot{B}(t) + 0.5B(t) = 30$$

$$k'(t)e^{-0.5t} + \cancel{k(t)e^{-0.5t}(-0.5)} + \cancel{0.5k(t)e^{-0.5t}} = 30$$

$$\Rightarrow \quad k'(t)e^{-0.5t} = 30$$

3.4) Separate

$$k'(t) = 30e^{0.5t}$$

Integrate both sides w.r.t. t:

$$\int k'(t) = \int 30e^{0.5t}\,dt$$

$$K(t) = \frac{30}{0.5}e^{0.5t} + C = 60e^{0.5t} + C$$

3.5 Plug $K(t)$ back into homogenous solution

$$B_h(t) = Ke^{-0.5t}$$

$$B(t) = K(t)e^{-0.5t}$$

$$\Rightarrow \quad B(t) = 60 - Ce^{-0.5t} \quad \leftarrow \text{General soln}$$

Complete soln: $\quad B(0) = 0$

$$0 = 60 + Ce^{0}$$

$$C + 60 = 0$$

$$C = -60$$

$$\boxed{B(t) = 60 - 60e^{-0.5t}} \quad \leftarrow \text{Complete}$$

$\Rightarrow$ Work through:

$$K(t) = 60e^{0.5t} + C = 60e^{0.5t} - 60$$

$$K(t)e^{-0.5t} = (60e^{0.5t} - 60)e^{-0.5t}$$

$$= 60 - 60e^{-0.5t}$$